Drop Ceiling — Step-by-Step Complexity Expansion

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How the system grew from 12 sliders to a self-learning light organism. Each step adds one new concept on top of the previous one. Diagrams use Mermaid syntax.

> For the full architectural reference, see BEHAVIOR_DIAGRAMS.md.

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1 — Control 12 DMX Values Directly with 12 Sliders

The initial step was to test the hardware and software configuration to confirm that the LED panels could be controlled in real time by the software.

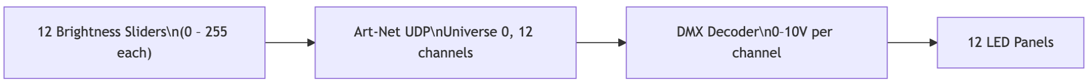

Diagram 1

Inputs

Input	Type	Range
Brightness per panel	Slider × 12	0 – 255

Output

- 12 DMX values (0 – 255) → Art-Net → DMX Decoder → 0 – 10V per panel

Key Parameters

Parameter	Value
DMX channels	12 (1 per panel)
Art-Net target	10.42.0.200
Universe	0

2 — Virtual Point Light

After testing several methods (wave fields, spring physics, radial gradients, vector controllers), a spatial control system emerged: a **virtual point light** that illuminates each panel based on the light's brightness and distance to the panel. A virtual sensor at the center of each panel reads the value based on the initial light value and the **falloff distance**. This meant 12 independent channels could be controlled with just 3 spatial parameters.

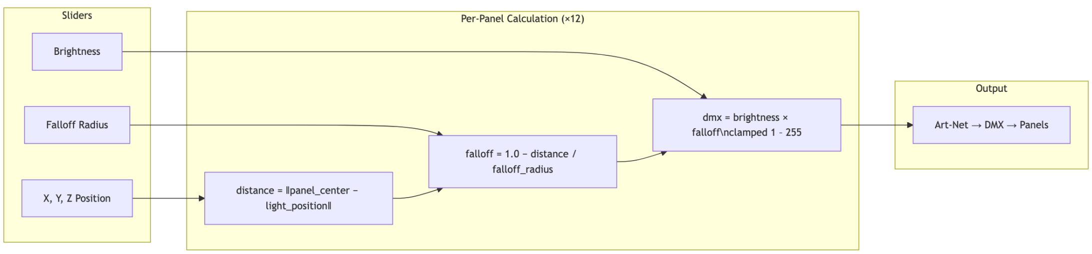

Diagram 2

Inputs

Input	Type	Range
Light X position	Slider	−290 to −30 cm
Light Y position	Slider	0 to 150 cm
Light Z position	Slider	−32 to 28 cm
Brightness	Slider	0 – 255
Falloff Radius	Slider	20 – 200 cm

Output

- 12 DMX values computed from distance-based falloff

Key Parameters

Parameter	Value
Panel layout	4 units × 3 panels
Unit spacing	80 cm center-to-center
Unit X positions	−30, −110, −190, −270 cm
Panel Y positions	90 cm (top), 30 cm (lower two)
Panel Z positions	0 cm (top), ±12 cm (lower, angled 22.5°)

What This Unlocked

Changing from 12 independent sliders to a single spatial light source meant the output was **spatially coherent** — panels near the light were always brighter than panels far away, without manual coordination.

---

3 — Animated Point Light Controller

The next stage moved from **direct manual control** to **animated movement** within a fixed area. Instead of positioning sliders, the user clicks to set a target position and the light lerps toward it. A **pulse cycle** (sine wave) was added to make the light breathe between a min and max brightness.

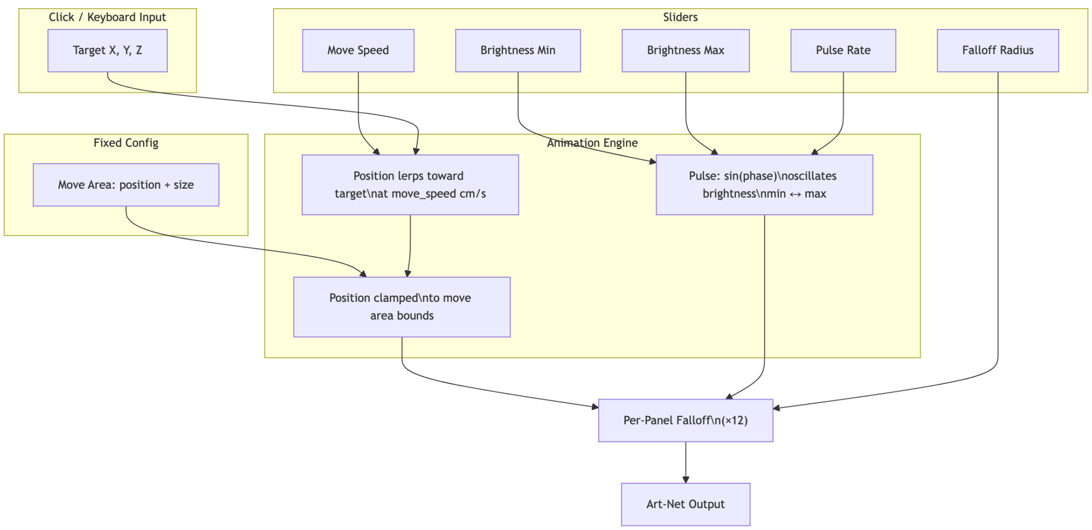

Diagram 3

Inputs

Input	Type	Range
Move Speed	Slider	5 – 100 cm/s
Brightness Min	Slider	0 – 255
Brightness Max	Slider	0 – 255
Pulse Rate	Slider	500 – 8000 ms
Falloff Radius	Slider	20 – 200 cm
Target X, Y, Z	Click / Arrow keys	Within move area
Move Area	Fixed config	Position + Size (X, Y, Z)

Output

- Animated light position + pulsing brightness → 12 DMX values

Key Parameters

Parameter	Value
Default move area	260 × 150 × 60 cm
Pulse method	`sin(phase)` oscillating 0.0 – 1.0
Brightness equation	`brightness_min + pulse × (brightness_max − brightness_min)`
Position interpolation	Linear lerp at `move_speed` cm/s

What This Unlocked

The light was now **alive** — it moved on its own, breathed, and had a spatial boundary. But it had no awareness of people.

---

4 — OSC Input: Pedestrian Data

The first connection between **real-world tracking** and the light. A pedestrian simulator (and later real cameras) sends person positions over OSC. The light now receives X, Z coordinates for each tracked person.

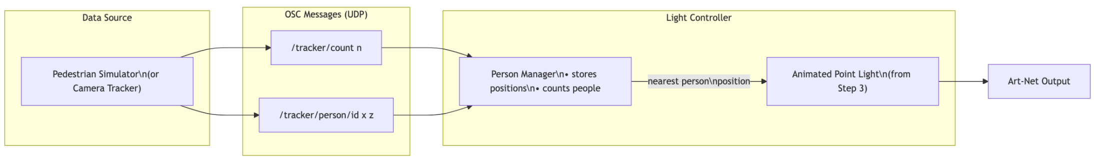

Diagram 4

Inputs

Input	Type	Range
`/tracker/count`	OSC int	0 – N
`/tracker/person/<id>`	OSC float × 2	X (cm), Z (cm)
All Step 3 sliders	Slider	same as above

Output

- Same as Step 3, but light target now follows person positions

Key Parameters

Parameter	Value
OSC port	7000
Protocol	UDP
Update rate	~25 Hz
Simulator pedestrian types	PASSIVE (walk-through), ACTIVE (wander), CURIOUS (enter-and-explore)

What This Unlocked

The light could now **respond to people**. But it treated every person the same — no concept of engagement, zones, or behavioral modes.

---

5 — Zone Classification: Active vs Passive

People are now classified into **two zones** based on their Z coordinate. The **active zone** is directly under the panels where people are "engaging." The **passive zone** is the sidewalk traffic beyond. This single classification drives all future behavioral decisions.

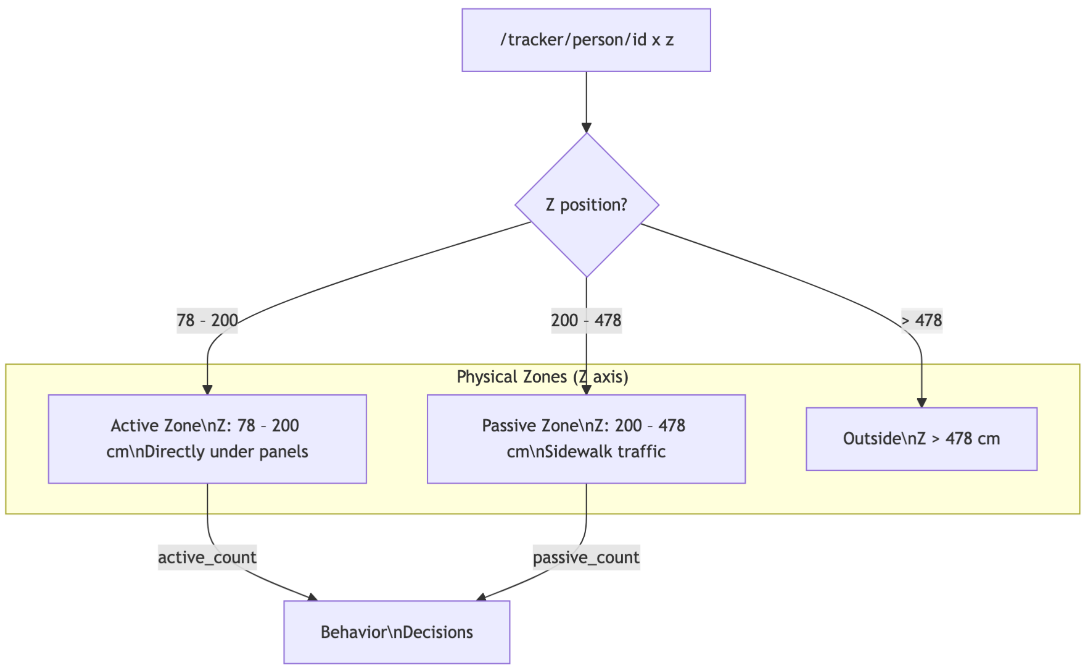

Diagram 5

Inputs

Input	Type
Person X, Z	OSC float × 2

Key Parameters

Parameter	Value
Active zone depth	~120 cm (directly under panels)
Passive zone depth	~270 cm (sidewalk beyond)
Zone boundaries	Loaded from `world_coordinates.json`

What This Unlocked

The system now knows the **difference between someone standing under the installation and someone walking past on the sidewalk**. This is the foundation of all behavioral modes.

---

6 — Four Behavioral Modes

The light is always in one of four modes based on zone occupancy. Each mode defines a **base personality** — how fast the light moves, how bright it shines, and whether it follows someone or wanders on its own.

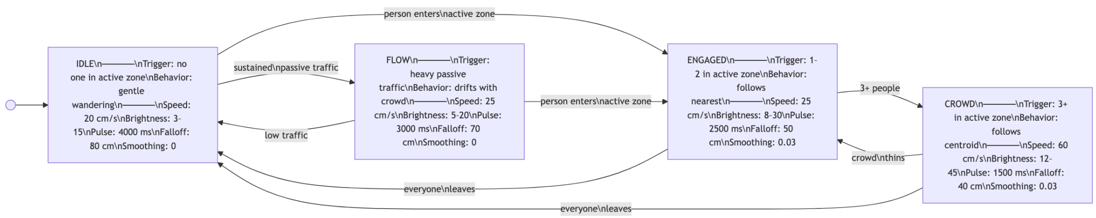

Diagram 6

Mode Base Parameters

Parameter	IDLE	ENGAGED	CROWD	FLOW
`move_speed`	20 cm/s	25 cm/s	60 cm/s	25 cm/s
`brightness_min`	3	8	12	5
`brightness_max`	15	30	45	20
`pulse_speed`	4000 ms	2500 ms	1500 ms	3000 ms
`falloff_radius`	80 cm	50 cm	40 cm	70 cm
`follow_smoothing`	0	0.03	0.03	0
`wander_interval`	5.0 s	4.0 s	0.0 s	3.0 s

What This Unlocked

The light now has **distinct behaviors** depending on who's around. An empty sidewalk gets a gentle wander; a person standing under the panels gets direct attention; a group gets high energy. The transition between modes is the first step toward the light feeling intentional.

7 — Mode Transition Smoothing

Raw mode switching causes **jarring jumps** — brightness snapping from 15 to 45 in a single frame. This step adds **stickiness timers** (conditions must persist before switching) and **transition interpolation** (parameters lerp from old to new values over time).

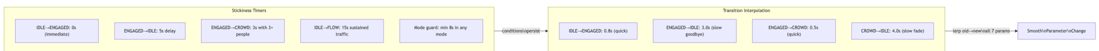

Diagram 7

Key Parameters

Transition	Stickiness	Interpolation
IDLE → ENGAGED	0 s (immediate)	0.8 s
ENGAGED → IDLE	5 s	3.0 s
ENGAGED → CROWD	3 s	0.5 s
CROWD → ENGAGED	5 s	2.0 s
CROWD → IDLE	5 s	4.0 s
IDLE → FLOW	15 s	2.0 s
FLOW → IDLE	10 s	3.0 s
FLOW → ENGAGED	0 s (immediate)	0.8 s
Min mode duration	8 s (any mode)	—

Design Choice

Engaging is fast (0.5–0.8 s), disengaging is slow (2.0–4.0 s). The light is eager to connect and reluctant to let go.

8 — Wander Box: Spatial Constraint

Instead of roaming freely, the light picks random targets inside a **3D bounding volume** (the wander box) and lerps toward them. In IDLE, the box spans the full panel array. In ENGAGED, the box **contracts and anchors around the person**. All future modifiers work by **reshaping the box**, not by steering the light directly.

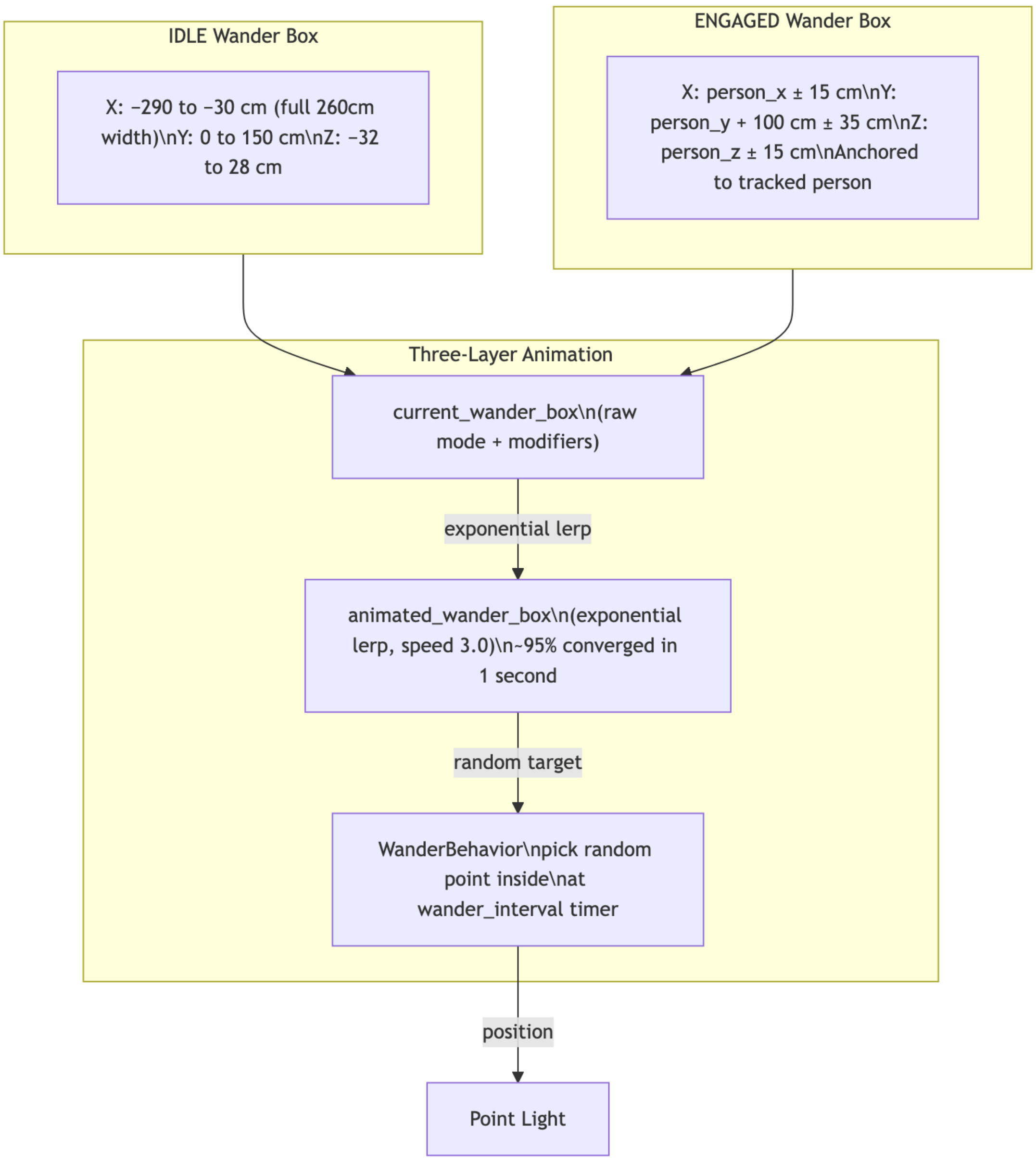

Diagram 8

Key Parameters

Parameter	Value
IDLE box width	260 cm (full array)
ENGAGED contraction	±15 cm around person
2 people weighting	70/30 toward nearest
3+ people	Centroid of all positions
Animation lerp speed	3.0 (exponential, ~1s converge)
Wander interval (IDLE)	5.0 s
Wander interval (ENGAGED)	4.0 s
Position lerp	3% per frame

What This Unlocked

The light's movement now has **spatial intent**. In IDLE it drifts broadly; in ENGAGED it orbits the person. The same wander mechanism drives both — only the box dimensions change.

---

9 — Gesture System (10 Types)

Short animations that interrupt the normal wander pattern to give the light **expressive texture**. Gestures fire based on conditions (someone entering, someone leaving, boredom) and are weighted by cooldowns and context.

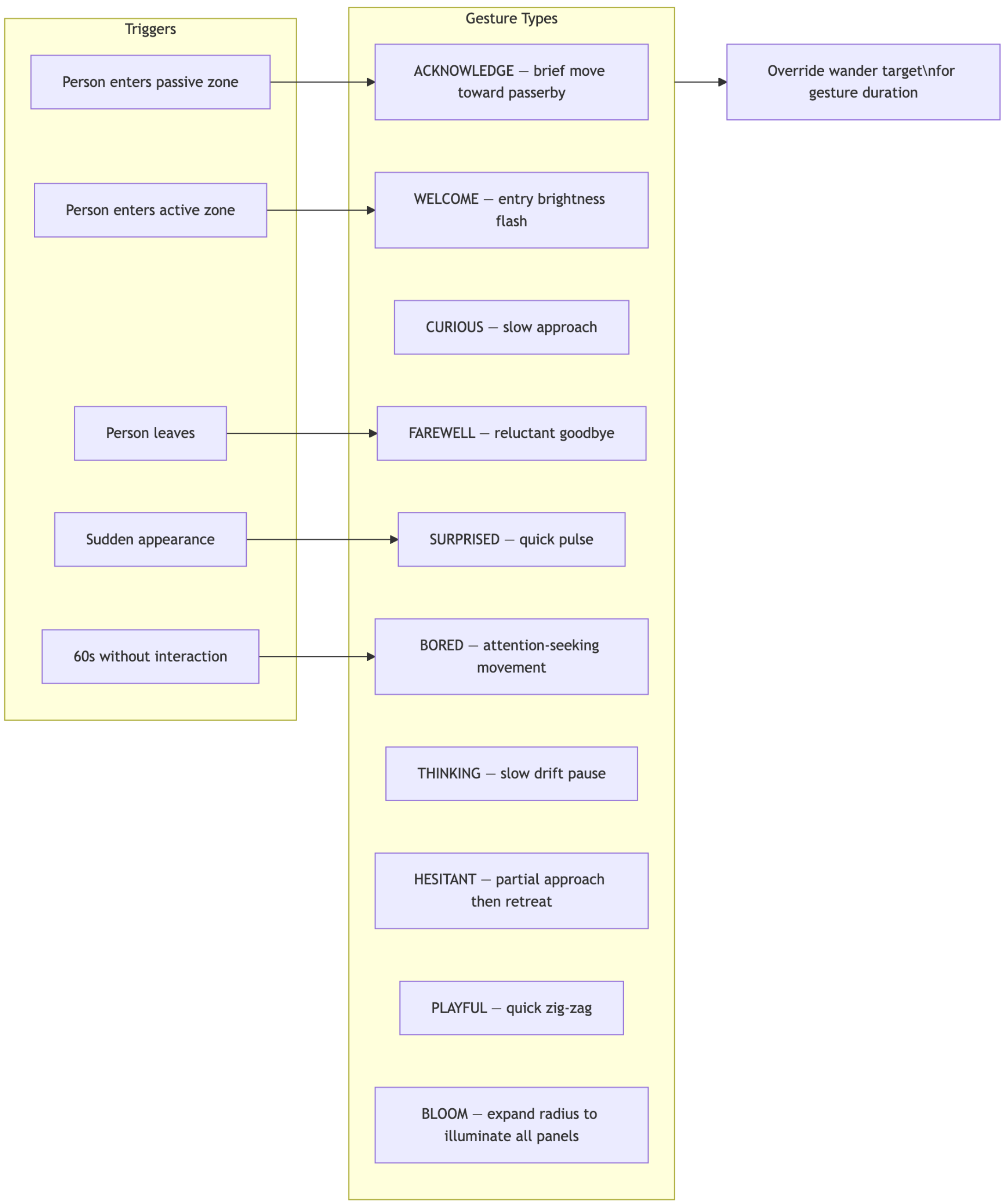

Diagram 9

Key Parameters

Parameter	Value
Bloom cooldown	45 s
Bloom duration	3.0 s
Bloom chance	~15% per minute
Bloom radius	300 cm (covers all panels)
Welcome flash	+25 brightness, 0.8 s
Boredom threshold	60 s without interaction

What This Unlocked

The light can now **express itself** — welcoming newcomers, seeking attention when bored, saying reluctant goodbyes. These are short punctuations on top of the continuous mode behavior.

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10 — Personality System: 6 MetaParameter Sliders

A layer of abstraction above the mode base values. Six personality sliders (0.0 – 1.0) define the light's **character**, and six global multipliers scale the final output. Together they transform the hard-coded mode values into something tuneable and expressive.

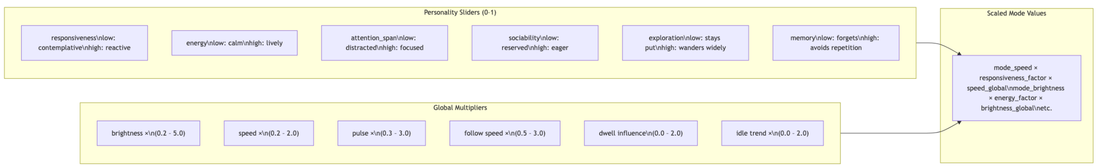

Diagram 10

How Personality Maps to Output

Slider	Affects	Low (0.0)	High (1.0)
responsiveness	`move_speed`, `follow_smoothing`	×0.6 (slow)	×1.4 (fast)
energy	`pulse_speed`, `brightness`	×1.3 period (slow), ×0.7 bright	×0.7 period (fast), ×1.3 bright
attention_span	gesture selection	distracted, short gestures	focused, SETTLE gestures
sociability	gesture frequency	×1.5 interval (rare)	×0.5 interval (frequent)
exploration	`wander_interval`	×1.5 (slow picks)	×0.5 (fast picks)
memory	anti-repetition strength	0 (no suppression)	1.0 (strong suppression)

Example Calculation

With `responsiveness = 0.8` and `speed_global = 1.2` in IDLE mode: - Base speed: 20 cm/s - Responsiveness factor: lerp(0.6, 1.4, 0.8) = **1.24** - Final: 20 × 1.24 × 1.2 = **29.8 cm/s**

11 — Time-of-Day Modifiers

The light's behavior shifts based on the hour of day. A financial district location means rush hours have commuters who won't stop, mid-day has explorers, and nighttime is dead. The system scales brightness and pulse accordingly.

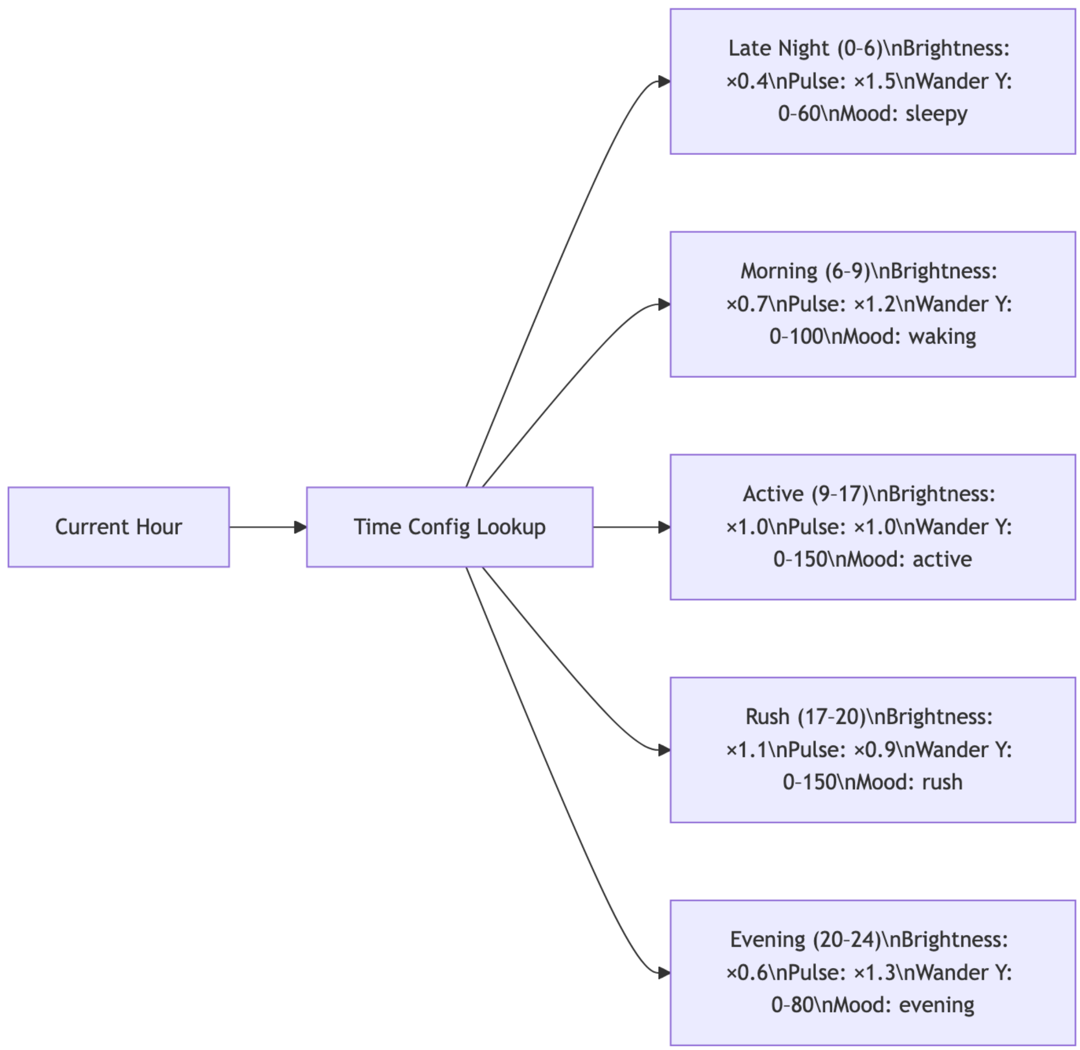

Diagram 11

Key Parameters

Period	Hours	Brightness ×	Pulse ×	Max Wander Y
Late night	0 – 6	0.4	1.5	60 cm
Morning	6 – 9	0.7	1.2	100 cm
Active	9 – 17	1.0	1.0	150 cm
Rush	17 – 20	1.1	0.9	150 cm
Evening	20 – 24	0.6	1.3	80 cm

12 — Dwell Phase System

When someone stays in the active zone, the interaction deepens through four progressive **dwell phases**. Each phase increases the light’s intimacy and commitment.

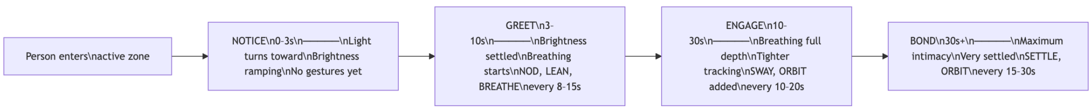

Diagram 12

Dwell Phase Parameters

Phase	Time	Breathing Depth	Gesture Interval	New Gestures
Notice	0 – 3 s	0 (none)	none	—
Greet	3 – 10 s	0.4 (subtle)	8 – 15 s	NOD, LEAN, BREATHE
Engage	10 – 30 s	0.7	10 – 20 s	+ SWAY, ORBIT
Bond	30 s+	1.0 (full)	15 – 30 s	+ SETTLE

Dwell Bonus

Brightness and follow smoothing increase with dwell time, scaled by the `dwell_influence` global multiplier (default 1.0, range 0.0 – 2.0).

13 — Engaged Interaction Gestures + Breathing Overlay

Six new gesture types for **ongoing engagement** (added on top of the original 10), plus a continuous **breathing overlay** — a slow sine wave that modulates brightness and falloff radius to make the light feel alive.

6 Engaged Gestures

Gesture	Duration	Motion	Phase-Gated
NOD	1.0 – 1.2 s	Small Y bob (12 cm)	Greet+
LEAN	1.5 – 1.8 s	Brief Z shift toward person (15 cm)	Greet+
SWAY	3.0 – 4.0 s	Gentle lateral X oscillation (18 cm)	Engage+
ORBIT	4.0 – 5.0 s	Lazy circle (X: 15 cm, Y: 8 cm)	Engage+
SETTLE	3.0 s	Tighten Z by 8 cm, shrink radius by 10	Bond
BREATHE	4.0 – 6.0 s	Visible brightness wave	Greet+

Breathing Overlay Parameters

Parameter	Value
Ramp-up time	8.0 s (0 → full depth)
Base period	6.0 s per breath cycle
Brightness modulation	±12% at full depth
Falloff radius modulation	±6% at full depth
Phase rate	1.047 rad/s (one cycle per 6 s)
Type	Multiplicative sine wave

What This Unlocked

The light now feels like it **breathes with you**. The deeper the engagement, the more synchronized the rhythm. Gestures add punctuation — a nod of acknowledgment, a gentle lean in, a comfortable sway.

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14 — Camera Tracking Integration

The pedestrian simulator is replaced (or supplemented) by **real YOLO-based camera tracking**. Two Reolink PoE cameras capture RTSP video, YOLO 11n detects people, ArUco calibration projects pixel positions to floor coordinates, and cross-camera fusion produces stable tracks.

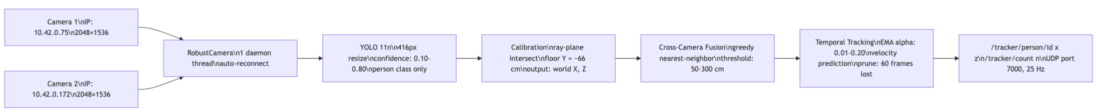

Diagram 13

Key Parameters

Parameter	Value
Cameras	2× Reolink RLC-520A
YOLO model	YOLO 11n
Detection resize	416 px wide
Confidence range	0.10 – 0.80 (slider)
Calibration method	ArUco markers (7) → solvePnP
Fusion threshold	50 – 300 cm (slider)
EMA smoothing alpha	0.01 – 0.20 (slider)
Track prune threshold	60 frames unseen
Output rate	25 Hz

Live Tuning Sliders (3)

Slider	What It Does	Range
Confidence	YOLO detection threshold	0.10 – 0.80
Fusion Distance	Max merge distance across cameras	50 – 300 cm
EMA Alpha	Track smoothing (low = smooth, high = responsive)	0.01 – 0.20

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15 — Proximity Response

A Z-distance-based modifier that changes the light's character when someone stands **close to the panels** vs **far away** at the edge of the active zone. Close means slower, brighter, more precise. Far means faster, dimmer, looser.

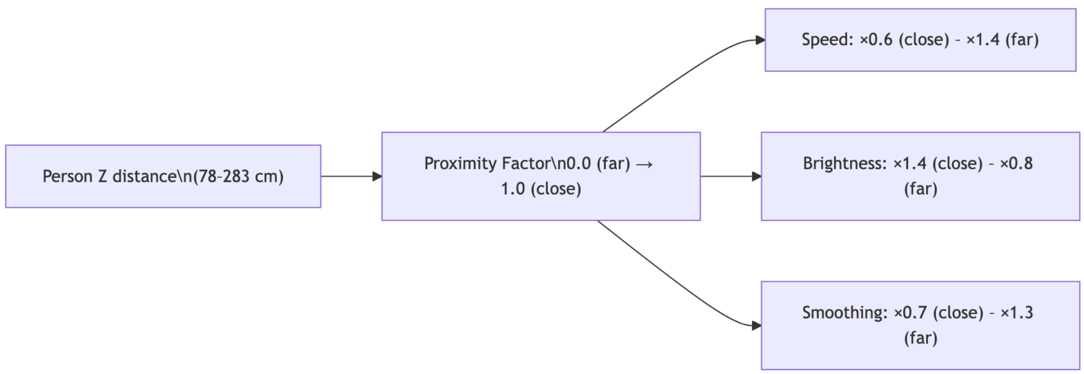

Diagram 14

Key Parameters

Parameter	Near (Z ≤ 100 cm)	Far (Z ≥ 280 cm)
Speed multiplier	×0.6	×1.4
Brightness multiplier	×1.4	×0.8
Smoothing multiplier	×0.7	×1.3
Interpolation	Linear between near and far	

16 — Trend Analysis: Multi-Timescale Windows

The system now tracks activity across **four rolling time windows** simultaneously. This gives the light awareness of not just “who’s here now” but “how busy has it been?”

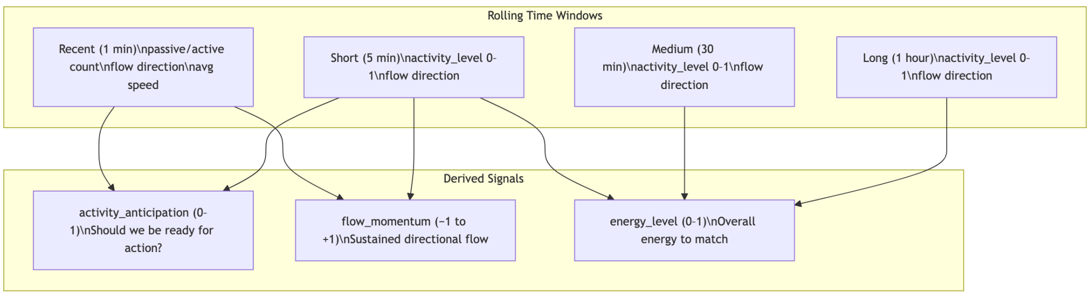

Diagram 15

Key Parameters

Window	Duration	Updates	Primary Use
Recent	1 min	continuous	Immediate reactivity
Short	5 min	continuous	Auto-tuning primary signal
Medium	30 min	continuous	General activity level
Long	1 hour	continuous	Big-picture energy

What This Unlocked

The light now has **temporal awareness**. A quiet afternoon after a busy morning feels different than a quiet afternoon after a quiet morning.

17 — Flow Tracking: Directional Awareness

A fast-updating tracker that measures **which direction pedestrians are moving** through the passive zone. This shifts the wander box toward incoming traffic, creating directional light emphasis even when nobody is actively engaged.

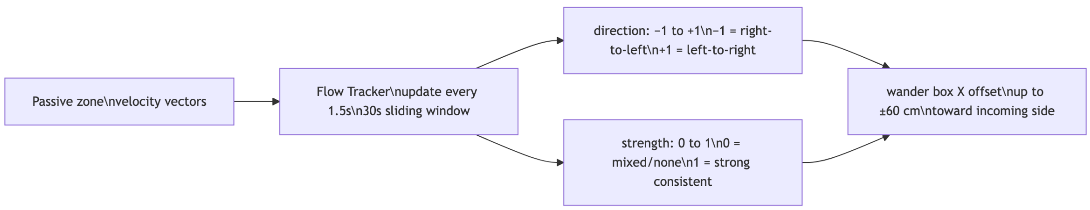

Diagram 16

Key Parameters

Parameter	Value
Update interval	1.5 s
Window duration	30 s sliding
EMA alpha	0.25 (responsive)
Max X offset	±60 cm
Direction	−1 (right-to-left) to +1 (left-to-right)

18 — Aggression System

A 0–1 value that **rises when the light hasn't engaged anyone for a while**. High aggression causes wider, more active wandering to attract attention. Capped by time-of-day (low during rush hours when commuters won't stop).

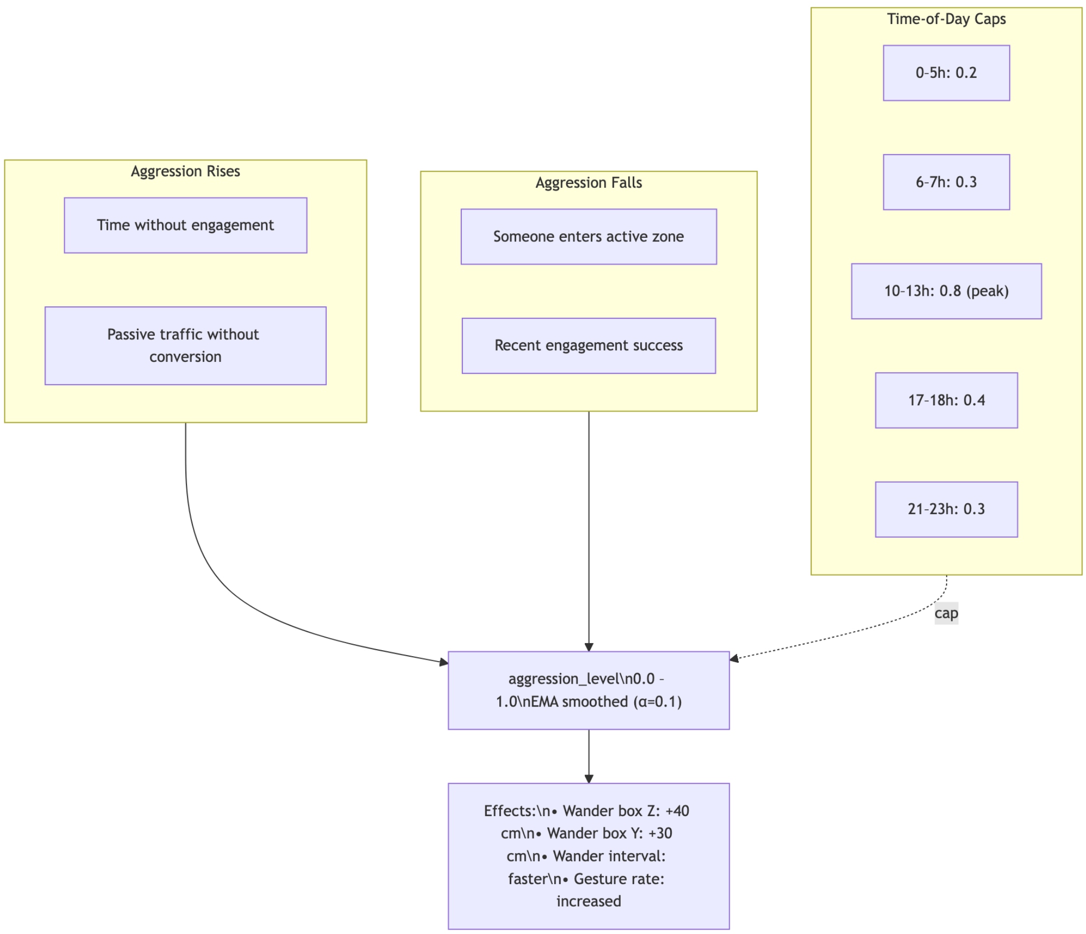

Diagram 17

Key Parameters

Parameter	Value
EMA alpha	0.1
Range	0.0 – 1.0
Peak cap (lunch: 10–13h)	0.8
Low cap (night: 0–5h)	0.2
Effects at high aggression	+40 cm Z wander, +30 cm Y wander, faster gestures

19 — Entry Pulse + People-Count Scaling

Two simple but impactful additions: a **brightness flash** when someone enters the active zone, and **per-person brightness scaling** in ENGAGED and CROWD modes.

Entry Pulse

Parameter	Value
Brightness boost	+25 DMX (one-shot)
Duration	0.8 s
Trigger	Person crosses from passive → active zone

People-Count Scaling

Parameter	Value
Brightness increase	+20% per person in active zone
Applies to	`brightness_min` and `brightness_max`

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20 — Anti-Repetition (Memory)

The `memory` personality slider scales an **anti-repetition system** that tracks recent gestures and positions, suppressing patterns that have been used recently. This prevents the light from falling into visible loops.

Key Parameters

Parameter	Value
Memory slider range	0.0 – 1.0
Effect at 1.0	Full suppression of recently used gestures
Effect at 0.0	No suppression (may repeat)
Tracking	Recent gesture types + recent position history

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21 — Almost-Engaged Attraction

Detects people who **slow down in the passive zone** near the active zone boundary — prime targets for attraction. Three strategies are A/B tested to learn which works best.

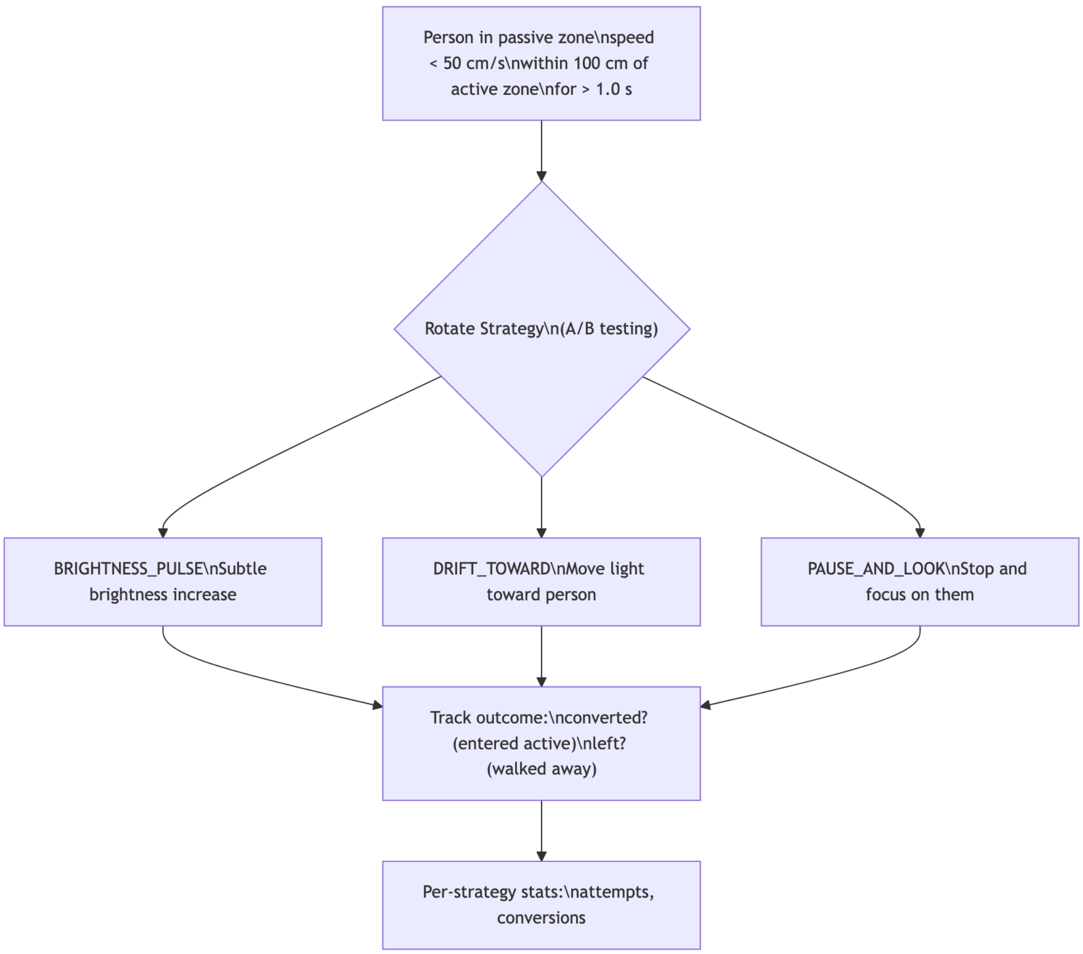

Diagram 18

Key Parameters

Parameter	Value
Slow speed threshold	< 50 cm/s
Near active threshold	Within 100 cm
Min detection time	1.0 s
Attraction cooldown	5.0 s between attempts
Drift X offset	±50 cm toward candidate

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22 — Feedback Learning

A slow learning system that logs **what the light was doing when people engaged** and gradually weights successful behaviors higher. Creates emergent learning: “What was I doing when people engaged? Do more of that.”

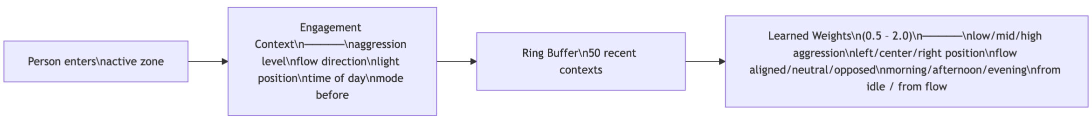

Diagram 19

Key Parameters

Parameter	Value
Buffer size	50 engagement contexts
Learning rate	±0.02 per engagement
Weight range	0.5 – 2.0
Dimensions	aggression × position × flow × time × mode
Effect	Weights multiply corresponding behavior parameters

23 — AutoTuning Feedback Loop

The `AutoTuningManager` runs every 5 seconds and **continuously adjusts all 12 personality/global parameters** based on observed activity. This creates a light that learns and adapts its character over hours.

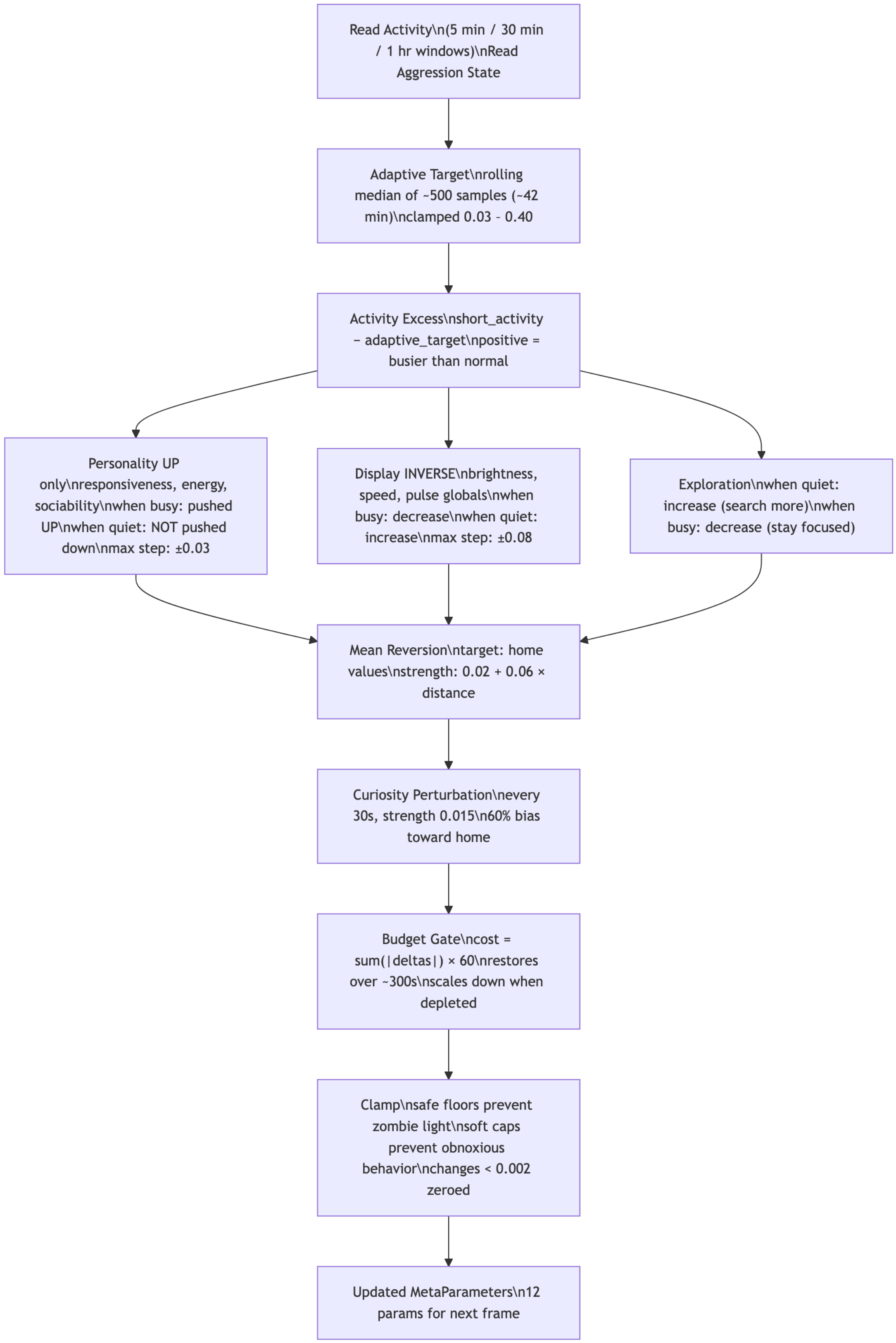
Read Activity\n(5 min / 30 min / 1 hr windows)\nRead Aggression State
Adaptive Target\nrolling median of ~500 samples (~42 min)\nclamped 0.03 – 0.40
Activity Excess\nshort_activity − adaptive_target\npositive = busier than normal
Personality UP only\nresponsiveness, energy, sociability\nwhen busy: pushed UP\nwhen quiet: NOT pushed down\nmax step: ±0.03
Display INVERSE\nbrightness, speed, pulse globals\nwhen busy: decrease\nwhen quiet: increase\nmax step: ±0.08
Exploration\nwhen quiet: increase (search more)\nwhen busy: decrease (stay focused)
Mean Reversion\ntarget: home values\nstrength: 0.02 + 0.06 × distance
Curiosity Perturbation\nevery 30s, strength 0.015\n60% bias toward home
Budget Gate\ncost = sum(|deltas|) × 60\nrestores over ~300s\nscales down when depleted
Clamp\nsafe floors prevent zombie light\nsoft caps prevent obnoxious behavior\nchanges < 0.002 zeroed
Updated MetaParameters\n12 params for next frame

Diagram 20

AutoTuning Constraints

Parameter	Safe Floor	Soft Cap	Home Value
responsiveness	0.30	0.90	0.50
energy	0.25	0.85	0.45
attention_span	0.10	—	0.50
sociability	0.20	—	0.45
exploration	0.15	—	0.40
memory	—	—	0.30
brightness_global	0.60	3.0	1.20
speed_global	0.35	1.6	0.70
pulse_global	0.35	2.0	0.80
follow_speed_global	0.60	—	1.00
dwell_influence	—	—	0.50
idle_trend_weight	0.10	—	0.40

Key Design Choice: Asymmetric Adjustment

Personality sliders are only pushed **up** when activity is high — never pushed down. Only mean reversion gently brings them back toward home values. This prevents the light from becoming permanently suppressed on a quiet sidewalk.

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24 — Daily Learning + Reset Cycles

The system persists what it learns across days. At midnight, the current parameter state is snapshotted and blended into “learned home values” for the next startup. Four daily resets prevent long-term drift.

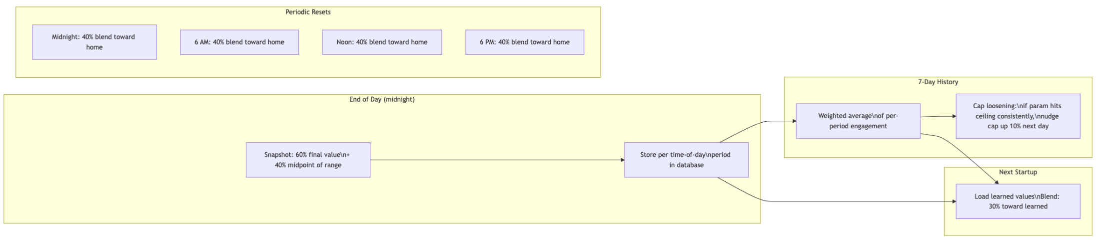


Diagram 21

Key Parameters

Parameter	Value
Daily snapshot blend	60% final + 40% midpoint
Startup blend	30% toward learned values
Reset hours	{0, 6, 12, 18}
Reset strength	40% toward home
Weekly history	7-day weighted average
Cap loosening	10% per day if consistently hit

25 — WebSocket Public Viewer

Real-time state is broadcast over WebSocket to a **Three.js web viewer** that renders the light, panels, tracked people, wander box, and track zones in 3D. Accessible remotely via Tailscale Funnel.

Diagram 22

Key Parameters

Parameter	Value
WebSocket port	8765
Broadcast rate	~15 FPS
Payload	JSON: light pos, brightness, panel DMX, people, mode
Auto-reconnect	3 s
Viewer engine	Three.js + OrbitControls
Hosting	GitHub Pages

26 — Production 24/7 Hardening

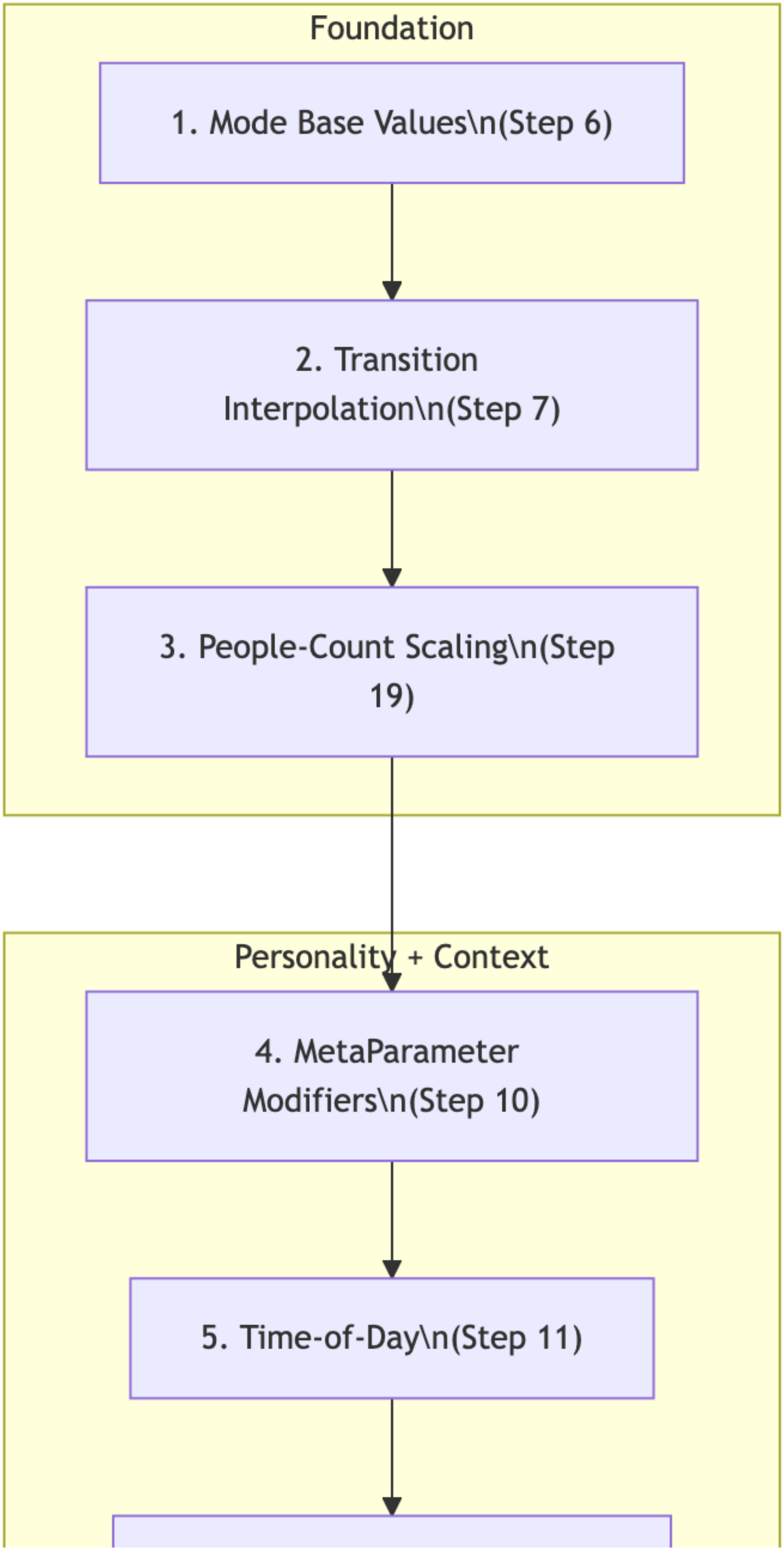
The final layer adds operational stability for continuous unattended operation: health monitoring, database pruning, single-instance locking, and persistent settings.

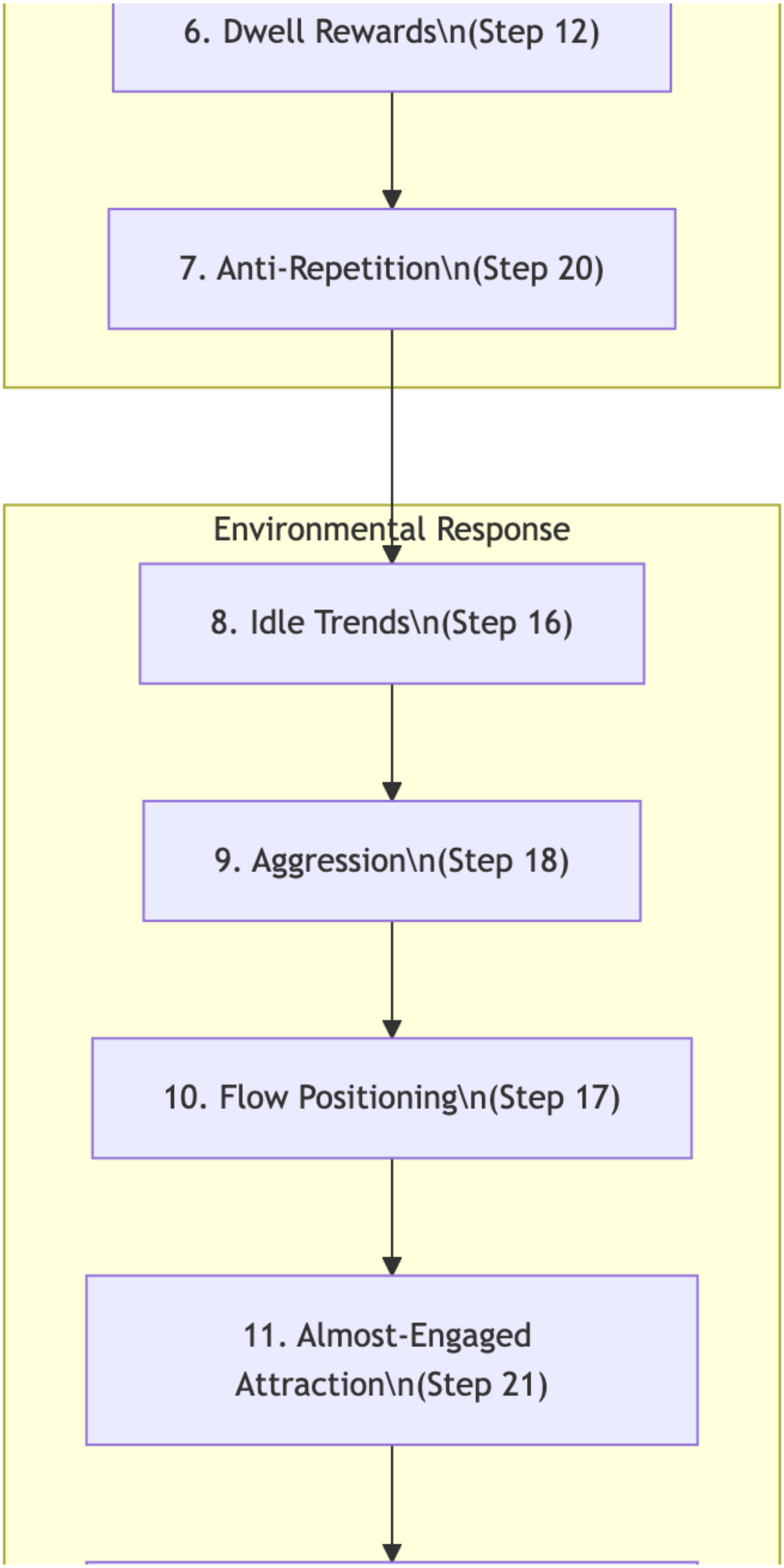
Key Parameters

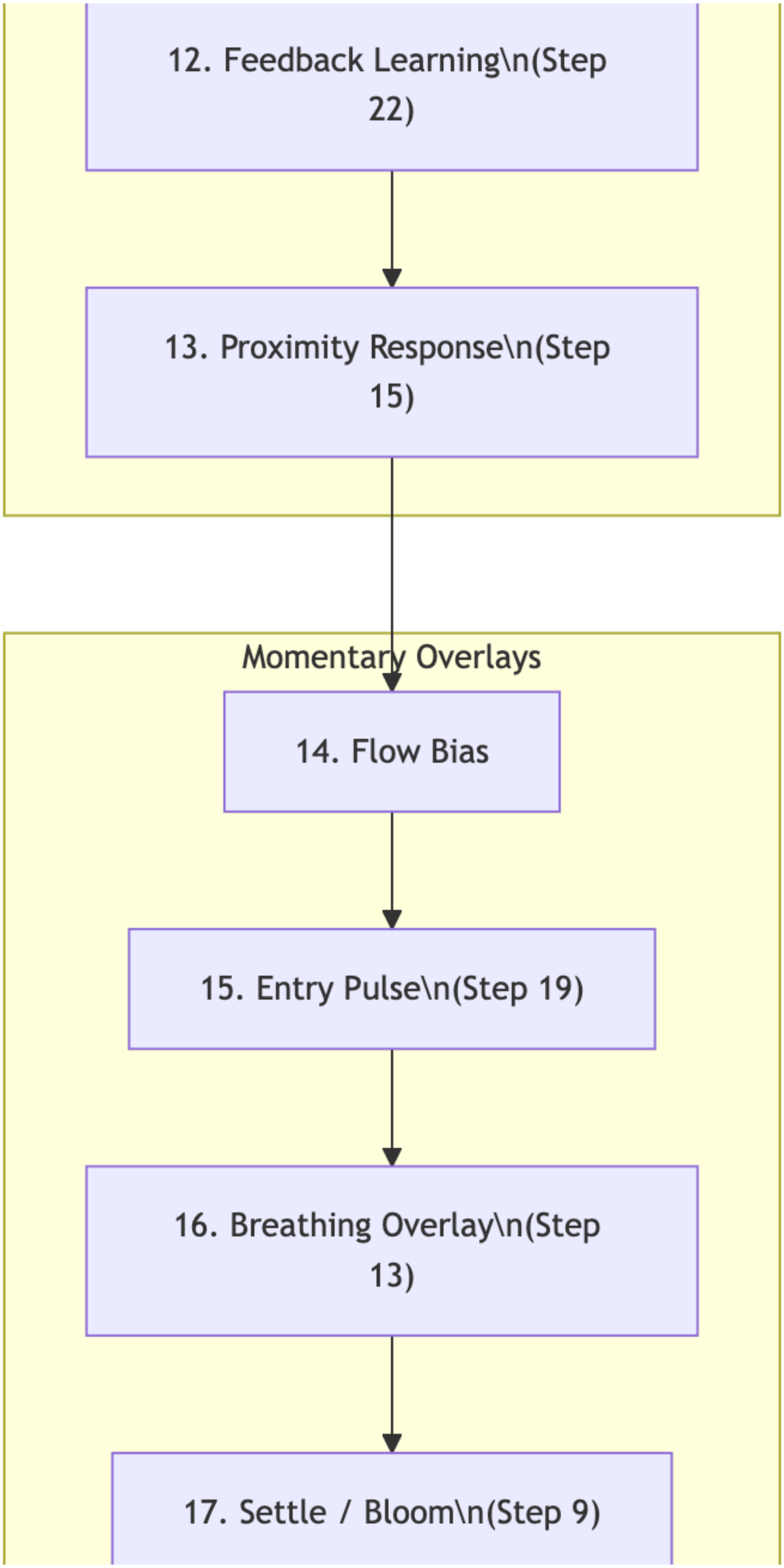
Parameter	Value
Health log interval	Every 5 minutes
Database pruning	Every hour
Raw event retention	48 hours (aggregated before deletion)
Single instance lock	`fcntl` file lock at `/tmp/lightController.lock`
Settings auto-save	Every 5 seconds
YOLO model reload	Every hour (prevent tracker drift)
Slider settings file	`slider_settings.json`
Override file	`autotune_overrides.json` (hot-reload every 30 s)

Summary: The Full 17-Layer Parameter Pipeline

Every frame, `calculate_parameters()` in `light_behavior.py` passes the output through **17 sequential layers** — the accumulated result of every step above:


Foundation
1. Mode Base Values\n(Step 6)
2. Transition Interpolation\n(Step 7)
3. People-Count Scaling\n(Step 19)
Personality + Context
4. MetaParameter Modifiers\n(Step 10)
5. Time-of-Day\n(Step 11)


6. Dwell Rewards\n(Step 12)
7. Anti-Repetition\n(Step 20)
Environmental Response
8. Idle Trends\n(Step 16)
9. Aggression\n(Step 18)
10. Flow Positioning\n(Step 17)
11. Almost-Engaged Attraction\n(Step 21)



12. Feedback Learning\n(Step 22)
13. Proximity Response\n(Step 15)
Momentary Overlays
14. Flow Bias
15. Entry Pulse\n(Step 19)
16. Breathing Overlay\n(Step 13)
17. Settle / Bloom\n(Step 9)

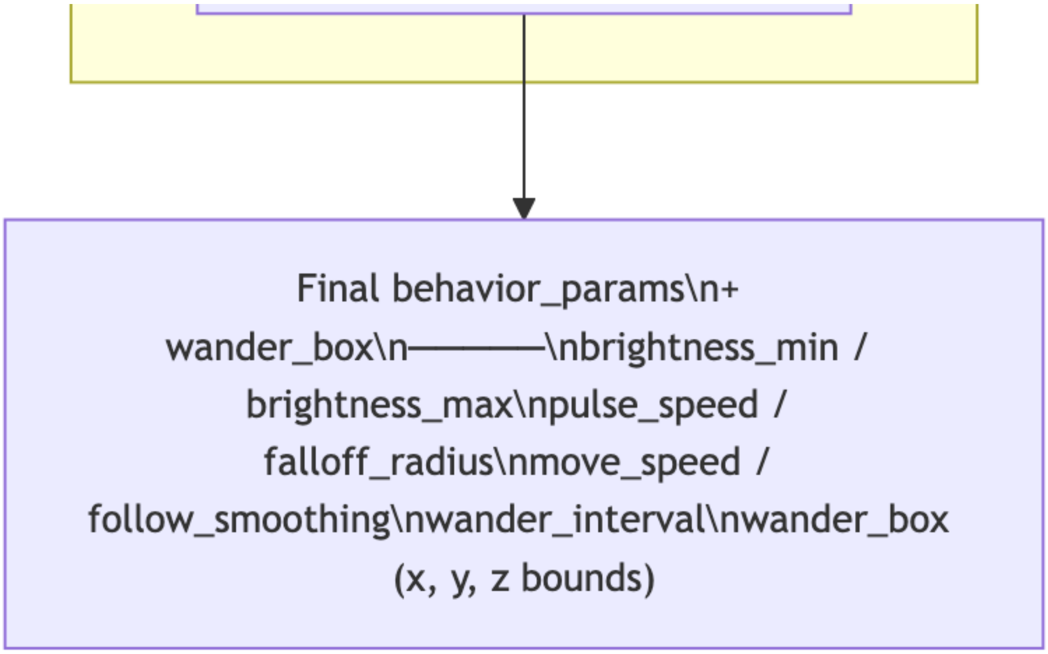

Final behavior_params\n+
wander_box\n————\nbrightness_min /
brightness_max\npulse_speed /
falloff_radius\nmove_speed /
follow_smoothing\nwander_interval\nwander_box
(x, y, z bounds)

Diagram 23

Complete Parameter Index

Every parameter in the system, in the order they were introduced:

Step	Parameter	Range / Unit	Purpose
1	DMX channel values	0 – 255	Raw panel brightness
2	Light X, Y, Z	cm	Virtual light position
2	Falloff Radius	20 – 200 cm	Distance-based panel illumination range
3	Move Speed	5 – 100 cm/s	Light movement rate
3	Brightness Min / Max	0 – 255	Pulse range
3	Pulse Rate	500 – 8000 ms	Breathing sine wave period
3	Move Area	cm^3	Position boundary
5	Active / Passive zone	Z thresholds	Person classification
6	Mode (IDLE/ENGAGED/CROWD/FLOW)	enum	Behavioral state
7	Stickiness timers	0 – 15 s	Delay before mode switch
7	Transition durations	0.5 – 4.0 s	Param interpolation time
7	Min mode duration	8 s	Anti-flicker guard
8	Wander Box (x, y, z min/max)	cm	Light movement boundary
8	Wander Interval	0 – 5 s	Target pick frequency
8	Animation lerp speed	3.0	Box convergence rate
9	Gesture types	10 (later 16)	Expressive animation set
9	Bloom cooldown	45 s	Min time between blooms
10	Personality sliders	0.0 – 1.0 × 6	Light character
10	Global multipliers	0.2 – 5.0 × 6	Output scaling
11	Time-of-day multipliers	per period	Brightness, pulse, wander Y

Step	Parameter	Range / Unit	Purpose
12	Dwell phases	4 thresholds	Engagement depth
12	Dwell influence	0.0 – 2.0	Bonus scaling
13	Breathing depth	0 – 1.0	Overlay intensity
13	Breathing period	6.0 s	Breath cycle
13	Engaged gesture amplitudes	8 – 18 cm	Per-gesture motion
14	YOLO confidence	0.10 – 0.80	Detection threshold
14	Fusion distance	50 – 300 cm	Cross-camera merge
14	EMA alpha	0.01 – 0.20	Track smoothing
15	Proximity Z near / far	100 / 280 cm	Speed/bright/smooth scaling
16	Trend windows	1m / 5m / 30m / 1h	Activity memory
17	Flow direction	–1 to +1	Crowd movement vector
17	Flow X offset	±60 cm	Wander box shift
18	Aggression level	0.0 – 1.0	Attention-seeking intensity
18	Time-of-day aggression caps	0.2 – 0.8	Max aggression by hour
19	Entry pulse boost	+25 DMX, 0.8 s	Welcome flash
21	Attraction strategies	3 types	A/B tested conversion
22	Feedback weights	0.5 – 2.0	Learned behavior multipliers
22	Learning rate	0.02	Weight change per event
23	AutoTune cycle	5 s	Adjustment frequency
23	Max step (personality)	±0.03	Per-cycle change limit

Step	Parameter	Range / Unit	Purpose
23	Max step (global)	±0.08	Per-cycle change limit
23	Safe floors	per param	Minimum viable values
23	Soft caps	per param	Maximum sane values
23	Home values	per param	Mean-reversion targets
23	Curiosity interval	30 s	Random exploration
23	Budget cost scale	60.0	Change rate limiter
24	Daily snapshot blend	60/40	Learning persistence
24	Startup blend	30%	Learned value incorporation
24	Reset hours	{0, 6, 12, 18}	Drift prevention
25	WebSocket rate	~15 FPS	Viewer update rate
26	Health log interval	300 s	Monitoring frequency
26	DB prune interval	3600 s	Cleanup frequency
26	DB retention	48 h	Raw event lifespan

Each step in this document corresponds to a conceptual addition. In practice, many of these were developed together or iterated on across versions (DEV → V1 → V2 → V2.5 → V3 → V4 → production). See BEHAVIOR_DIAGRAMS.md for the full architectural reference with all feedback loops visible.